

Eklavya Sharma

Curriculum Vitae

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in [sharmaeklavya2](#)  [sharmaeklavya2](#) 

Research Interests

Economics and computation, Social choice theory, Game theory, Markets, Approximation algorithms

Education

- Aug 2021 – **PhD**, *Department of Industrial & Enterprise Systems Engineering (ISE)*,
present *University of Illinois at Urbana-Champaign (UIUC)*, IL, USA
Research on data markets and fair allocation. Advised by Prof. Jugal Garg .
- July 2019 – **M.Tech. (Research)**, *Computer Science and Automation (CSA)*, Indian
July 2021 *Institute of Science (IISc)*, Bangalore, GPA: 9.7 / 10.0
Research on approximation algorithms for variants of bin packing and knapsack.
Advised by Prof. Arindam Khan .
- Aug 2014 – **B.E. (Hons) Computer Science**, *Birla Institute of Technology and Science*
June 2018 *(BITS)*, Pilani, India, GPA: 9.14 / 10.00

Publications

- Submitted to Equilibrium pricing for oligopolistic data markets
ICML'26 with Bhaskar Ray Chaudhury, Jugal Garg, and Jiaxin Song
- Submitted to Revenue-optimal pricing for budget-constrained buyers in data markets
EC'26 with Bhaskar Ray Chaudhury, Jugal Garg, and Jiaxin Song
- AAMAS'26 Exploring relations among fairness notions in discrete fair division 
with Jugal Garg
- AAMAS'26 Proportional and Pareto-optimal allocation of chores with subsidy 
with Jugal Garg and Xiaowei Wu
- EC'24 Improving approximation guarantees for maximin share  
with Hannaneh Akrami, Jugal Garg, and Setareh Taki
- IJCAI'23 New fairness concepts for allocating indivisible items  
with Ioannis Caragiannis, Jugal Garg, Nidhi Rathi, and Giovanna Varricchio
- IJCAI'23 Simplification and improvement of MMS approximation  
with Hannaneh Akrami, Jugal Garg, and Setareh Taki

- FSTTCS'23 Two-player matrix games repeated until collision [↗](#)
with Aniket Murhekar
- ArXiv Automating the search for small hard examples to approximation algorithms [↗](#)
- FSTTCS'22 Approximation algorithms for multidimensional packing [↗](#) [↗](#)
with Arindam Khan and K.V.N. Sreenivas
- Algorithmica, APPROX'21 Tight approximation algorithms for geometric bin packing with skewed items [↗](#) [↗](#) [↗](#), with Arindam Khan
- FSTTCS'21 Harmonic algorithms for packing d -dimensional cuboids into bins [↗](#) [↗](#)

Fellowships and Achievements

- Aug 2024 **Mavis Future Faculty Fellowship, UIUC** [↗](#)
A program that trains fellows on various aspects of an academic career through workshops, seminars, and activities.
- April 2024 **William A. Chittenden II Award, UIUC** [↗](#)
Given to an outstanding graduate student at ISE.
- July 2023 **Dr. MNS Swamy Medal for Best MTech (Research) Thesis, IISc** [↗](#)
- April 2023 **Sharp Outstanding Graduate Student Award, UIUC** [↗](#)
- Aug 2021 – July 2022 **Samuel Brainin Engineering Fellowship, UIUC**
- March 2018 **Graduate Aptitude Test in Engineering (GATE), India**
Rank 86 (out of $\sim 100,000$ candidates) in the 'Computer Science and IT' test.
- Aug 2014 – Dec 2015 **BITS Pilani Merit Scholarship**
Scored GPA among top 2% of students in the first three semesters of my Bachelors.

Invited Talks

- 27 Oct 2023 **Fair allocation of indivisible items**
Capital Area Theory Seminar, University of Maryland, College Park
- 22 Dec 2022 **Existence and computation of epistemic EFX allocations** [↗](#)
Indian Institute of Science, Bangalore

Professional Service

Program committee member for EC'26, AAMAS'26, AAI'26, EC'25, AAMAS'25, AAMAS'24.

Conference subreviewer for EC'24, ICALP'24, EC'23, ESA'23, IJCAI'23, FCT'23, STOC'22, SAGT'22, MFCS'21.

Journal reviewer for Optimization Letters (2024).

Implementation Projects

- April 2024 – **Exploring relations among fairness notions in discrete fair division**
Dec 2024 *Published in AAMAS'26, code at github.com/sharmaeklavya2/cpigjs*
Wrote a program in JavaScript that helped prove many new results for the problem of fairly allocating indivisible items among multiple agents. These results help arrange fairness notions in a hierarchy.
- Sept 2023 – **Automating the search for hard examples to approximation algorithms.** *code: github.com/sharmaeklavya2/code2dtree, paper in IJCAI'23.*
March 2024 Wrote a python library that converts any function to a decision tree (for a fixed input size). Then wrote a program to find tight hard examples for approximation algorithms by running a linear program for each leaf of the algorithm's decision tree. Used this to find tight hard examples for the then-best-known approximate-MMS algorithm for fairly allocating goods.
- May 2020 **Interactive app for rectangular bin packing**
code: github.com/sharmaeklavya2/packing-game

Work Experience

- Teaching Assistant, IE 300: Analysis of Data** (Fall'22, Spring'24, Spring'25, Spring'26), UIUC
- Spring 2023 **Teaching Assistant, IE 310: Deterministic models in optimization**, UIUC
- Fall 2020 **Teaching Assistant, Design and Analysis of Algorithms**, IISc Bangalore
- Aug 2018 – **Software Engineer, media.net**, Bangalore, India
- July 2019 *Topics: machine learning, large-scale systems.*
media.net is an advertisement-technology company. I worked on improving their machine-learning-based algorithm for bidding in real-time ad auctions.
- Jan 2018 – **Intern** , *American Express*, Gurgaon, India
- June 2018 *Topics: neural networks, machine learning, big data.*
Trained a neural network to predict credit-card defaulting. The input format was unconventional, so I devised a custom architecture. Its performance was at par with the production model, which was tuned over many years.
- May 2017 – **Intern, Directi**, Mumbai, India
- July 2017 *Topics: machine learning.*
Made Directi's news article classification algorithm recognize more categories.
- May 2016 – **Google Summer of Code (GSoC) Student** , *Zulip*
- Aug 2016 *Topics: software development.*
Zulip is an open-source group chat application. 3 students were selected from over 100 applicants to work on Zulip as part of the GSoC program.
- Annotated python code (~ 50,000 lines) for use with a static type-checker.
 - Migrated code to Python 3 by switching to newer dependencies, using automated code conversion, standardizing string types, and fixing bugs.

Computer Skills

ℒ_AT_EX, Python, HTML, CSS, JavaScript, C/C++, Java, SQL.

Selected Coursework

UIUC:

- (CS 580) Algorithmic Game Theory: grade A
- (CS 598 TH1) Recent Advances in Theoretical CS: grade A+
- (CS 473) Algorithms: grade A+
- (IE 511) Integer Programming: grade A
- (IE 519) Combinatorial Optimization: grade A
- (IE 410) Advanced Stochastic Processes and Applications: grade A+
- (IE 411) Optimization of Large Systems: grade A+

IISc Bangalore:

- Approximation Algorithms: grade A+
- Design and Analysis of Algorithms: grade A+
- Computational Methods of Optimization: grade A+
- Cryptography: grade A